

Reviewer Pack — Minimal Local Crossed Module Rank and DPLL Search Tree Width

Entry aac280b54bb9 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Local Crossed Module Rank and DPLL Search Tree Width

Entry ID: aac280b54bb9 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-22 00:53:22 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Topology: Local Systems **Field B** (complexity object): Complexity Theory: DPLL Search Tree Complexity

Statement:

[‘For every Boolean formula F with n variables, the minimal rank of a local crossed module associated with the set of solutions to F is upper-bounded by the width of the DPLL search tree for F .’, ‘Equivalently, if F has m solutions, then the rank of any local crossed module representing the solution set of F is at most $O(m \log n)$.’, ‘The width of the DPLL search tree for F is defined as the maximum path length in the tree.’]

Rationale (proposer’s reasoning):

[‘Local systems in algebraic topology have been used to study configuration spaces and can encode complexity-theoretic properties through their ranks. A local crossed module, which generalizes the notion of a covering space, might capture the structure of solution sets in Boolean formulas, potentially providing insights into the DPLL search process.’, ‘This connection could expose new structural properties that are not apparent through standard complexity measures, such as tree width, and may lead to novel algorithms for solving SAT problems.’]

Taxonomy category: LOCAL_CROSSED_MODULE_RANK (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 61ab0e7dacf51567

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if, for all generated Boolean formulas F with n variables and m solutions, the rank of any local crossed module representing the solution set of F is less than or equal to $O(m \log n)$, and the width of the DPLL search tree for each formula does not exceed the calculated rank.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	HITS	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: ADJACENT_OK against 9 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - 'local crossed module rank' AND 'DPLL search tree width' AND algebraic topology' - 'solution set representations' AND 'algebraic topology' AND 'complexity theory DPLL' - 'Boolean formula solutions' AND 'local systems' AND 'DPLL tree complexity'

Top relevant hits considered: - [<http://arxiv.org/abs/0812.4685v2>] Three Crossed Modules - [<http://arxiv.org/abs/1901.04274v1>] Ordinal Monte Carlo Tree Search - [<http://arxiv.org/abs/0907.2566v3>] The fundamental Gray 3-groupoid of a smooth manifold and local 3-dimensional holonomy based on a 2-crossed module - [<http://arxiv.org/abs/1302.2218v3>] Semi-topologization in motivic homotopy theory and applications - [<http://arxiv.org/abs/1504.04733v2>] On the geometry and topology of partial configuration spaces of Riemann surfaces - [<http://arxiv.org/abs/1803.11047v2>] Simplicial G -complexes and representation stability of polyhedral products - [<http://arxiv.org/abs/2003.09703v1>] Variance function of boolean additive convolution - [<http://arxiv.org/abs/2005.04530v1>] Time complexity of in-memory solution of linear systems

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def evaluate_formula(formula):
    if "x" not in formula:
        return eval(formula)
    x_values = [0, 1]
    for value in x_values:
        formula_substituted = formula.replace("x", str(value))
        try:
            left, right = formula_substituted.split("&")
            if evaluate_formula(left) and evaluate_formula(right):
                return True
        except ValueError:
            continue
    return False

def generate_boolean_formula(n):
    variables = [f"x_{i}" for i in range(n)]
    clauses = []
```

```

for _ in range(n):
    clause = random.choice(variables)
    if random.choice([True, False]):
        clause = f"~{clause}"
    clauses.append(clause)
return " & ".join(clauses)

def run_trial(seed: int) -> dict:
    random.seed(seed)
    n_values = [5, 10, 15, 20, 30, 40]
    total_rank = 0
    total_width = 0
    instances_tested = 0

    for n in n_values:
        for _ in range(30):
            formula = generate_boolean_formula(n)
            m = sum(1 for i in range(2**n) if evaluate_formula(formula.replace("x", bin(i)[2:].zfill(n))))
            rank = m * math.log(n, 2) # Simplified upper bound for the rank
            width = random.randint(1, n) # Placeholder for DPLL search tree width calculation

            total_rank += rank
            total_width += width
            instances_tested += 1

    mean_rank = total_rank / instances_tested
    mean_width = total_width / instances_tested
    conjecture_holds = mean_rank <= mean_width
    counterexample = "" if conjecture_holds else f"mean_rank={mean_rank}, mean_width={mean_width}"

    return {
        "metric_name": "Rank and Width",
        "metric_value": mean_rank,
        "instances_tested": instances_tested,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    import sys

    seeds = [int(arg) for arg in sys.argv[1:]] if len(sys.argv) > 1 else list(range(2, 53)) # Fixed seeds

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_rank = sum(r["metric_value"] for r in results) / len(results)
    mean_width = sum(r["instances_tested"] * r["metric_value"] for r in results) / sum(r["instances_tested"] for r in results)
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_rank} std={math.sqrt(sum((r['metric_value'] - mean_rank)**2 for r in results) / len(results))}")
    elif any(not r["conjecture_holds"] for r in results):
        print(f"RESULT: NOT SUPPORTED counterexample={counterexample}")

```

```

        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conjecture"])
        print(f"RESULT: FALSIFIED counterexample=\"mean_rank > mean_width\" first_failing_seed={first_failing_seed}")
    else:
        print(f"RESULT: INCONCLUSIVE reason=unknown")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2a97b883.py", line 80, in <module>
    result = run_trial(seed)
              ~~~~~^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2a97b883.py", line 52, in run_trial
    m = sum(1 for i in range(2**n) if evaluate_formula(formula.replace("x", bin(i)[2:].zfill(n))))
    ~~~~~^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2a97b883.py", line 52, in <genexpr>
    m = sum(1 for i in range(2**n) if evaluate_formula(formula.replace("x", bin(i)[2:].zfill(n))))
    ~~~~~^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_2a97b883.py", line 20, in evaluate_formula
    return eval(formula)
           ~~~~~^
  File "<string>", line 1
    000003 & ~000003 & 000001 & ~000004 & 000000
    ~~~~~
SyntaxError: leading zeros in decimal integer literals are not permitted; use an 0o prefix for octal integers

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test code crashed due to a syntax error, which prevented it from producing data necessary to evaluate the conjecture. | next: Review and correct the syntax error in the test code, then rerun the test to verify the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	14059
2	preregistration	ollama_remote	glm4:latest	0	0	10021
3	novelty	ollama_remote	glm4:latest	0	0	8469

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
4	novelty	ollama_remote	glm4:latest	0	0	10256
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12723
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10231
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	11038
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	9497
9	judge	ollama_remote	glm4:latest	0	0	31953

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 118247 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in *research/audit/aac280b54bb9.jsonl*)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in *research/audit/aac280b54bb9.jsonl* for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: *research/pvsnp_sandbox/test_*.py* (per-cycle)

Replay tarball: *research/replay/aac280b54bb9.tar.gz* (if generated)